

Case Studies

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[Practical]

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions

```
harshika@Harshika:~$ mkdir linux_task4
harshika@Harshika:~$ cd linux_task4
harshika@Harshika:~/linux_task4$ touch public.txt private.txt script.sh
harshika@Harshika:~/linux_task4$ nano public.txt
harshika@Harshika:~/linux_task4$ nano private.txt
harshika@Harshika:~/linux_task4$ nano script.sh
harshika@Harshika:~/linux_task4$ ls -l
total 12
-rw-r--r-- 1 harshika harshika 24 Aug  6 17:45 private.txt
-rw-r--r-- 1 harshika harshika 23 Aug  6 17:45 public.txt
-rw-r--r-- 1 harshika harshika 31 Aug  6 17:46 script.sh
harshika@Harshika:~/linux_task4$ chmod 600 private.txt
harshika@Harshika:~/linux_task4$ chmod 644 public.txt
harshika@Harshika:~/linux_task4$ chmod +x script.sh
harshika@Harshika:~/linux_task4$ ls -l
total 12
-rw----- 1 harshika harshika 24 Aug  6 17:45 private.txt
-rw-r--r-- 1 harshika harshika 23 Aug  6 17:45 public.txt
-rwxr-xr-x 1 harshika harshika 31 Aug  6 17:46 script.sh
harshika@Harshika:~/linux_task4$ ./script.sh
Hello Linux
harshika@Harshika:~/linux_task4$ ls -l
total 12
-rw----- 1 harshika harshika 24 Aug  6 17:45 private.txt
-rw-r--r-- 1 harshika harshika 23 Aug  6 17:45 public.txt
-rwxr-xr-x 1 harshika harshika 31 Aug  6 17:46 script.sh
harshika@Harshika:~/linux_task4$ chmod -x script.sh
harshika@Harshika:~/linux_task4$ ls -l
total 12
-rw----- 1 harshika harshika 24 Aug  6 17:45 private.txt
-rw-r--r-- 1 harshika harshika 23 Aug  6 17:45 public.txt
-rw-r--r-- 1 harshika harshika 31 Aug  6 17:46 script.sh
harshika@Harshika:~/linux_task4$ |
```

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories.

```
harshika@Harshika:~$ mkdir permission_lab
harshika@Harshika:~$ cd permission_lab
harshika@Harshika:~/permission_lab$ touch test.txt
harshika@Harshika:~/permission_lab$ nano test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ chmod u-r test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
--w-r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ chmod u+r test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ chmod u-w test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
-r--r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ nano test.txt
harshika@Harshika:~/permission_lab$ chmod u+w test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
harshika@Harshika:~/permission_lab$ |
```

Task 8: Backup and Restore

Objective: Create backups and restore files using cp.

```

harshika@Harshika:~$ mkdir linux_task6
harshika@Harshika:~$ cd linux_task6
harshika@Harshika:~/linux_task6$ touch important.txt
harshika@Harshika:~/linux_task6$ nano important.txt
harshika@Harshika:~/linux_task6$ mkdir backup
harshika@Harshika:~/linux_task6$ cp important.txt backup/
harshika@Harshika:~/linux_task6$ tree
.
├── backup
│   └── important.txt
└── important.txt

2 directories, 2 files
harshika@Harshika:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
harshika@Harshika:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important.txt

2 directories, 2 files
harshika@Harshika:~/linux_task6$ rm important.txt
harshika@Harshika:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└──

2 directories, 1 file
harshika@Harshika:~/linux_task6$ ls
backup
harshika@Harshika:~/linux_task6$ cp backup/important_backup.txt .
harshika@Harshika:~/linux_task6$ ls
backup  important_backup.txt
harshika@Harshika:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

2 directories, 2 files
harshika@Harshika:~/linux_task6$ mv important_backup.txt important_restored.txt

```

```

harshika@Harshika:~/linux_task6$ mv important_backup.txt important_restored.txt
harshika@Harshika:~/linux_task6$ cat important_restored.txt
Operating System Lab
Linux Commands
Backup and Restore
Learning cp command
Learning mv command
Learning chmod command
Learning mkdir command
Learning rm command
Learning directories
End of File
harshika@Harshika:~/linux_task6$ |

```


Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together

```
harshika@Harshika:~$ mkdir linux_project
harshika@Harshika:~$ cd linux_project
harshika@Harshika:~/linux_project$ mkdir source backup docs
harshika@Harshika:~/linux_project$ touch source/main.c source/helper.c source/README.txt
harshika@Harshika:~/linux_project$ ls source/
README.txt  helper.c  main.c
harshika@Harshika:~/linux_project$ nano source/README.txt
harshika@Harshika:~/linux_project$ head -5 source/README.txt
Operating System Lab
Linux Commands
File Management
Directory Management
Permissions
harshika@Harshika:~/linux_project$ tail -5 source/README.txt
Backup Folder
Docs Folder
Learning Commands
Operating Systems
End of File
harshika@Harshika:~/linux_project$ tree
.
├── backup
├── docs
└── source
    ├── README.txt
    ├── helper.c
    └── main.c

4 directories, 3 files
harshika@Harshika:~/linux_project$ cp source/main.c backup/
harshika@Harshika:~/linux_project$ ls backup/
main.c
harshika@Harshika:~/linux_project$ ls
backup docs source
harshika@Harshika:~/linux_project$ cp source/README.txt docs/
harshika@Harshika:~/linux_project$ ls docs
README.txt
harshika@Harshika:~/linux_project$ mv source/helper.c source/functions.c
harshika@Harshika:~/linux_project$ ls source
README.txt  functions.c  main.c
harshika@Harshika:~/linux_project$ chmod 644 source/README.txt
```

```
harshika@Harshika:~/linux_project$ chmod 644 source/README.txt
harshika@Harshika:~/linux_project$ ls -l source/README.txt
-rw-r--r-- 1 harshika harshika 311 Aug  7 03:14 source/README.txt
harshika@Harshika:~/linux_project$ rm backup/main.c
harshika@Harshika:~/linux_project$ ls backup
harshika@Harshika:~/linux_project$ rmdir backup/
harshika@Harshika:~/linux_project$ ls
docs source
harshika@Harshika:~/linux_project$ tree
.
├── docs
│   └── README.txt
└── source
    ├── README.txt
    ├── functions.c
    └── main.c

3 directories, 4 files
```

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file management exercise.

```
harshika@Harshika:~/linux_project$ mkdir student_portal
harshika@Harshika:~/linux_project$ cd student_portal
harshika@Harshika:~/linux_project/student_portal$ mkdir src docs backup logs config
harshika@Harshika:~/linux_project/student_portal$ ls
backup  config  docs  logs  src
harshika@Harshika:~/linux_project/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
harshika@Harshika:~/linux_project/student_portal$ tree
.
├── backup
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

6 directories, 5 files
harshika@Harshika:~/linux_project/student_portal$ nano logs/system.log
harshika@Harshika:~/linux_project/student_portal$ head -5 logs/system.log
System Started
User Login
Configuration Loaded
Database Connected
Service Started
harshika@Harshika:~/linux_project/student_portal$ tail -5 logs/system.log
Permissions Verified
Files Synced
System Healthy
System Shutdown
harshika@Harshika:~/linux_project/student_portal$ nano logs/system.log
harshika@Harshika:~/linux_project/student_portal$ tail -5 logs/system.log
Memory Checked
Permissions Verified
Files Synced
System Healthy
System Shutdown
harshika@Harshika:~/linux_project/student_portal$ cp docs/README.txt backup/
```

```
harshika@Harshika:~/linux_project/student_portal$ cp docs/README.txt backup/
harshika@Harshika:~/linux_project/student_portal$ ls backup
README.txt
harshika@Harshika:~/linux_project/student_portal$ mv backup/README.txt backup/README_backup.txt
harshika@Harshika:~/linux_project/student_portal$ ls backup
README_backup.txt
harshika@Harshika:~/linux_project/student_portal$ chmod 600 config/config.txt
harshika@Harshika:~/linux_project/student_portal$ ls -l config
total 0
-rw----- 1 harshika harshika 0 Aug  7 03:25 config.txt
harshika@Harshika:~/linux_project/student_portal$ chmod 644 docs/README.txt
harshika@Harshika:~/linux_project/student_portal$ ls -l docs
total 0
-rw-r--r-- 1 harshika harshika 0 Aug  7 03:25 README.txt
harshika@Harshika:~/linux_project/student_portal$ rm backup/README_backup.txt
harshika@Harshika:~/linux_project/student_portal$ ls backup
harshika@Harshika:~/linux_project/student_portal$ rmdir backup
harshika@Harshika:~/linux_project/student_portal$ ls
config  docs  logs  src
harshika@Harshika:~/linux_project/student_portal$ tree
.
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

5 directories, 5 files
harshika@Harshika:~/linux_project/student_portal$ |
```